

- [c.8] Kang, S., Kim, G., **Hwang, S.**, Park, J., Elsharkawy, A., and Kim, S. “Flip-Pelt: Motor-Driven Peltier Elements for Rapid Thermal Stimulation and Congruent Pressure Feedback in Virtual Reality.” *Proceedings of the 2024 ACM Symposium on User Interface Software and Technology*
- [c.7] Kim, G., **Hwang, S.**, Seong, M., Yeo, D., Rus, D., and Kim, S. “TimelyTale: A Multimodal Dataset Assessing Passenger's Demands for Explanations in Highly Automated Vehicles.” *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, Vol. 8, No.3
- [c.6] Jo, T., Yeo, D., Kim, G., **Hwang, S.**, and Kim, S. “WatchCap: Improving Scanning Efficiency in People with Low Vision through Compensatory Head Movement Stimulation.” *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, Vol. 8, No. 2
- [j.2] Lee, J., **Hwang, S.**, Kim, K., and Kim, S. “Evaluation of Visual, Auditory, and Olfactory Stimulus-Based Attractors for Intermittent Reorientation in Virtual Reality Locomotion.” *Virtual Reality* 28, 104 (2024)
- [c.5]  [Honorable Mention] **Hwang, S.**, Oh, J., Kang, S., Seong, M., Elsharkawy, A., and Kim, S. “ErgoPulse: Electrifying Your Lower Body With Biomechanical Simulation-based Electrical Muscle Stimulation Haptic System in Virtual Reality.” *Proceedings of the 2024 CHI conference on Human Factors in Computing Systems*
- [c.4]  [Honorable Mention] Elsharkawy, A., Ataya, A., Yeo, D., An, E., **Hwang, S.**, and Kim, S. “SYNC-VR: Synchronizing Your Senses to Conquer Motion Sickness for Enriching In-Vehicle Virtual Reality.” *Proceedings of the 2024 CHI conference on Human Factors in Computing Systems*
- [j.1] Lee, J., **Hwang, S.**, Ataya, A., and Kim, S. “Effect of Optical Flow and User VR Familiarity on Curvature Gain Thresholds for Redirected Walking.” *Virtual Reality* 28, 35 (2024)
- [c.3]  [Honorable Mention] **Hwang, S.**, Kim, Y., Seo, Y., and Kim, S. “Enhancing Seamless Walking in Virtual Reality: Application of Bone-Conduction Vibration in Redirected Walking.” *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*
- [c.2]  [Honorable Mention] Kang, Y., Choi, S., An, E., **Hwang, S.**, and Kim, S. “Designing Virtual Agent Human–Machine Interfaces Depending on the Communication and Anthropomorphism Levels in Augmented Reality.” *Proceedings of the 2023 International Conference on Automotive UI*
- [c.1] **Hwang, S.**, Lee, J., Kim, Y., Seo, Y., and Kim, S. “Electrical, Vibrational, and Cooling Stimuli-Based Redirected Walking: Comparison of Various Vestibular Stimulation-Based Redirected Walking Systems.” *Proceedings of the 2023 CHI conference on Human Factors in Computing Systems*

POSTERS & WORKSHOPS

- [w.4] **Hwang, S.**, Kang, S., Oh, J., Park, J., Shin, S., Luo, Y., DelPreto, J., Matusik, W., Rus, D., and Kim, S. “Proposal of a Framework for Enhancing Teleoperation Experience with Biomechanical Simulation-Based Electrical Muscle Stimulation in Virtual Reality.” *UbiComp '24: Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing*
- [w.3] Elsharkawy, A., Ataya, A., Yeo, D., Seong, M., **Hwang, S.**, DelPreto, J., Matusik, W., Rus, D., and Kim, S. “Adaptive In-Vehicle Virtual Reality for Reducing Motion Sickness: Manipulating Passenger Posture During Driving Events.” *UbiComp '24: Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing*
- [w.2] Choi, Y., Yeo, D., **Hwang, S.**, Seong, M., Moon, J., Luo, Y., Matusik, W., Rus, D., and Kim, K. “Intelligence Walker: A Seamless Mobility Assist Device for the Elderly.” *2024 IEEE ICRA Workshop on Wearable*
- [w.1] Kang, S., Kim, G., **Hwang, S.**, Park, J., Elsharkawy, A., and Kim, S. “Dual-sided Peltier Elements for Rapid Thermal Feedback in Wearables.” *2024 IEEE ICRA Workshop on Wearable*
- [p.4] Kim, Y., **Hwang, S.**, Oh, J., and Kim, S. “GaitWay: Gait Data-Based VR Locomotion Prediction System Robust to Visual Distraction.” *Extended Abstracts of the 2024 CHI conference on Human Factors in Computing Systems*
- [p.3] Gim, B., Kang, S., Kim, G., Yeo, D., **Hwang, S.**, and Kim, S. “Curving the Virtual Route: Applying Redirected Steering Gains for Active Locomotion in In-Car VR.” *Extended Abstracts of the 2024 CHI conference on Human Factors in Computing Systems*
- [p.2] **Hwang, S.**, Lee, J., Kim, Y., and Kim, S. “REVES: Redirection Enhancement Using Four-Pole Vestibular Electrode Stimulation.” *Extended Abstracts of the 2022 CHI conference on Human Factors in Computing Systems*
- [p.1] Lee, J., **Hwang, S.**, Kim, K., and Kim, S. “Auditory and Olfactory Stimuli-Based Attractors to Induce Reorientation in Virtual Reality Forward Redirected Walking.” *Extended Abstracts of the 2022 CHI conference on Human Factors in Computing Systems*

PATENTS & COPYRIGHTED CONTENTS

- [pa.2] Kim, S., Kang, S., Kim, G., **Hwang, S.**, Park, J., Elsharkawy, A. “Haptic wearable devices for tactile experiences.” *KR Patent App. KR102541073*
- [pa.1] **Hwang, S.**, Lee, J., Kim, Y., Seo, Y., and Kim, S. “Method and System for Supporting Walking in Virtual Environment.” *US Patent App. US20250159408A1 | KR Patent App. KR20250069327A*
- [cc.1, 2] Kim, S., Kang, S., Kang, Y., Kim, K., Seong, M., An, E., Yang, H., Yeo, D., Oh, J., Jeon, H., Jo, T., and **Hwang, S.** “Mobility-Linked Virtual Reality-Based Underwater Exploration Immersive Content Game Software ([cc.2]: Underwater Exploration & Ocean Trash Collection Game). & ([cc.1]: Underwater Exploration & Underwater Gem Collection Game).” *Copyright for Computer Program Works [cc.2]: C-2022-050134 & [cc.1]: C-2022-050133 (KR)*

INVITED TALKS

-  **University of Washington, Information School** | Invited lightning talk (25th anniversary Information School Research Fair) Mar 2026
-  **Toyota Research Institute** | Invited presentation (hosted by Alexandre Filipowicz) Aug 2025
-  **University of Chicago, Human Computer Integration Lab** | Invited presentation (hosted by Pedro Lopes) Jan 2024
-  **HCI Korea 2024, ACM SIGCHI** | Invited presentation on “Vestibular Stimuli-Based Redirected Walking” (hosted by Inseok Hwang) Jan 2024

ACADEMIC SERVICE

-  **Associate Chair**
 - UbiComp/ISWC 2024 Workshop on AI-infused Physical Systems (I4U)  **Service recognition for AC contributions**
-  **Conference Reviewer**
 - CHI (4 in 2026, 4 in 2025); CHI EA (1 in 2026, 2 in 2024, 1 in 2023); TEI (1 in 2025)  **Recognition for an outstanding review**; VRST (1 in 2024); UIST (1 in 2026, 1 in 2023)  **Recognition for an outstanding review**; EuroHaptics (2 in 2026); IMWUT (1 in 2026)
-  **Journal Reviewer**
 - International Journal of Human–Computer Interaction (3 in 2024, 5 in 2025, 5 in 2026; incl. multi-round reviews)

MEDIA COVERAGE

-  **“UW study finds touch screens in cars create a multitasking problem that impacts driving.”** Dec 2025
 - Outlets: UW News; GeekWire; KIRO Seattle's morning News; KOMO News; Spotlight on America; Techradar; El Confidential
-  **“Development of communication technology that informs road users of when autonomous vehicles will stop.”** Dec 2025
 - Outlets: Veritas Alpha; Kyosu Shinmun; AI Times

▣ “Expanding virtual space by 43%: Developing technology that overcomes VR walking limitations.” • Outlets: HelloDD; Veritas Alpha; Herald Business; News1 Korea; Newsworker	Dec 2025
▣ “Grandma, you no longer have to push a stroller: AI-equipped walking robot for older adults.” • Outlets: The Robot News; E-News Today; E-Dong-A; Newsworker; NEWSIS; AI Times; Seoul Economic Daily; Veritas Alpha	Jun 2025
▣ “Shaken car? No problem” – Enjoy VR without motion-sickness even while riding..” • Outlets: Veritas Alpha; OhmyNews; Namdo Ilbo; Herald Business; Web Economy	May 2025
▣ “Robot’s received force transmitted directly to human arm: Joint development by GIST and MIT.” • Outlets: Digital Chosun Ilbo; AI Times; Yonhap News Agency; EBN; News1 Korea; Newsworker; Veritas Alpha	May 2025
▣ “Autonomous driving safety information displayed at the right moment: GIST-MIT paper receives top 1% award.” • Outlets: NEWSIS; The University Journal; Newsworker; E-News Today; Veritas Alpha	Oct 2024
▣ “Wearable hat helps low-vision users by turning their heads more: WatchCap developed.” • Outlets: Brain Media; Medical News; ITDaily; Dong-A Science; Wide Economy; NEWSIS; The University Journal; and 3 more	Sep 2024
▣ “GIST research team receives two Honorable Mention awards at ACM CHI 2024 for VR interface work.” • Outlets: Namdo Ilbo; AI Times; ITDaily	May 2024
▣ “GIST virtual-reality interface research wins best-paper awards at IEEE ISMAR 2023.” • Outlets: BoanNews; AI Times; NEWSIS; News1 Korea; AVING News; Veritas Alpha; Newsworker; Newspim	Nov 2023